Application of Real Gas Flow Theory to Well Testing and Deliverability Forecasting

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ABSTRACT

Previous gas well test analyses have been based mainly upon linearizations of ideal gas flow results, although a method for drawdown analysis based upon real gas flow results has been proposed. Linearizations based upon ideal gas flow require estimation of gas physical properties at some sort of average pressure, and implicitly involve the assumption that pressure gradients are small everywhere in the reservoir. A new real gas flow equation has been developed by means of a substitution which couples pressure, viscosity and gas law deviation factor. This substitution has been called the real gas pseudo-pressure. Use of the real gas pseudo-pressure leads to simple equations describing real gas flow which do not contain pressure-dependent gas properties, and which do not require the assumption of small pressure gradients everywhere in the flow system.

Equations required to determine flow capacity, well condition and static formation pressure from pressure drawdown and build-up tests with the real gas pseudo-pressure concept are presented. Also shown are applications of the real gas pseudo-pressure to back-pressure testing, the gas materials balance and rigorous determination of average gas properties for previous gas flow equations. Included are sample calculations for well test analyses.

INTRODUCTION

A recent study[1] of the flow of the transient flow of real gases in ideal radial systems showed that it is possible to consider gas physical property dependence upon pressure by means of a substitution called the real gas pseudo-pressure. The substitution has the advantage that it enables engineering solutions for steady and transient flow of real gases that are more accurate and general than those previously available. The solutions are particularly important for the case of gas flow in tight, high-pressure formations under large drawdowns. Even for more ideal flow conditions, the real gas pseudo-pressure concept leads to important rules for finding average gas physical properties pertinent for older well test analyses.

Ref. 1 provides a detailed study of the theory behind the real gas flow correlations to be used in this paper. The purpose of this paper is to provide necessary engineering forms for use of the real gas flow results, and to illustrate applications.

The following considers flow of real gases in an ideal radial flow system. It is assumed that: (1) formation thickness, porosity, water saturation, absolute permeability, temperature and gas composition are constant; (2) gas compressibility and density are functions of pressure as described by the gas law $pv = znRT$; (3) gas viscosity is a function of pressure; (4) effective permeability to gas may be a function of pressure to account for liquid condensation; (5) condensate is immobile; and (6) the formation has no dip.

SUMMARY OF REAL GAS FLOW EQUATIONS

Combination of the continuity equation and Darcy's law for radial flow of a real gas yields the equation:

$$\frac{1}{r}\frac{\partial}{\partial r}\left[r\,\frac{p}{\mu(p)z(p)}\frac{\partial p}{\partial r}\right]=\frac{\phi}{k}\frac{\partial}{\partial t}\left[\frac{p}{z(p)}\right] \quad . \quad . \quad . \quad . \quad . \quad (1)$$

If the left-hand side of Eq. 1 is differentiated, a term involving $(\partial p^2/\partial r)^2$ arises which does not occur in ideal gas flow equations. Thus, modification of ideal gas flow solutions to fit real gas flow involves the assumption that pressure gradients are small. An alternate procedure is to make a change of variable in Eq. 1.

We define the real gas pseudo-pressure $m(p)$ as:

$$m(p) = 2\int_{p_m}^{p}\frac{p}{\mu(p)z(p)}dp \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (2)$$

As shown in Ref. 1, substitution of $m(p)$ in Eq. 1 leads to:

$$\frac{\partial^2 m(p)}{\partial r^2}+\frac{1}{r}\frac{\partial m(p)}{\partial r}=\frac{\phi\mu(p)c_g(p)}{k}\frac{\partial m(p)}{\partial t} \quad . \quad . \quad . \quad . \quad (3)$$

Eq. 3 looks like the diffusivity equation, but is still non-linear because the diffusivity depends upon pressure, or $m(p)$. Eq. 3 also closely resembles the equation for the flow of an ideal gas. Furthermore, the dependence of the diffusivity term upon $m(p)$ for the real gas is very similar to the dependence of the ideal gas diffusivity upon pressure squared. As is shown in Ref. 1, this leads immediately to practical solutions for Eq. 3 which are similar to the Aronofsky-Jenkins[2] ideal gas flow solutions. Engineering units will be used throughout the following. For production of a real gas in an ideal radial system at constant rate with a closed outer boundary:

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[1]References given at end of paper.

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$$\frac{1.987 \times 10^{-5}\, khTsc}{qp_{sc}T}\left[m(\bar{p})-m(p_w)\right] = \ln\frac{r_d}{r_w} + s + Dq \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (4)$$

where

$$\ln\frac{r_d}{r_w} = p_D(t_D) - 2\, t_D\left(\frac{r_w}{r_e}\right) \quad , \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (5)$$

and

$$t_D = \frac{0.000264\, kt}{\phi(\mu C_t)_i r_w^2} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (6)$$

The $m(\bar{p})$ is evaluated at an average pressure determined from a materials balance:

$$\left(\frac{p}{z}\right)_i - \left(\frac{\bar{p}}{z}\right) = \frac{Tp_{sc}G_p}{\pi h r_e^2 \phi S_g T_{sc}} = \int_{m(\bar{p})}^{m(p_i)} (\mu c_g)\, dm(p)$$

$$\approx (\mu c_g)_{avg}\,[m(p_i)-m(\bar{p})] \quad . \quad . \quad . \quad . \quad . \quad . \quad (7)$$

The $p_D(t_D)$ is the van Everdinger-Hurst[3] dimensionless pressure-drop function. For large ratios of r_e/r_w:

$$p_D(t_D) = \frac{1}{2}\,[\ln t_D + 0.80907] \quad , \text{ for } 100 \leqslant t_D \leqslant \frac{1}{4}\left(\frac{r_e}{r_w}\right)^2 \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (8)$$

$$p_D(t_D) = \ln\frac{r_e}{r_w} - \frac{3}{4} + 2\, t_D\left(\frac{r_w}{r_e}\right)^2 \quad , \text{ for } t_D \geqslant \frac{1}{4}\left(\frac{r_e}{r_w}\right)^2 \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (9)$$

Eqs. 4 through 9 are not an explicit solution to Eq. 3. But they do represent an excellent engineering approximation good for a wide range of conditions.

The total system compressibility in Eq. 6 (c_t) normally can be taken equal to the gas saturation-gas compressibility product S_gC_g. The porosity used is the *total* porosity.

In view of the finding that production of real gas can be linearized as above, we can write a very useful equation for the real gas pressure at the inner boundary of an infinite reservoir produced at constant rate, including a steady-state skin effect and non-Darcy flow term for $t_D > 100$, and before outer boundary effects are important:

$$m(p_{wf}) = m(p_i) - 5.792 \times 10^4 \frac{qp_{sc}T}{khT_{sc}}\left[\log_{10} t_D + 0.3513 + 0.87s + 0.87Dq\right] \quad . \quad . \quad . \quad . \quad . \quad (10)$$

If standard conditions are taken to be 14.7 psia and 60F:

$$m(p_{wf}) = m(p_i) - 1{,}637\frac{qT}{kh}\left[\log_{10} t_D + 0.3513 + 0.87s + 0.87Dq\right] . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (11)$$

And for long flowing times:

$$q = 1.987 \times 10^{-5}\frac{khT_{sc}\,[m(\bar{p})-m(p_{wf})]}{p_{sc}T\left[\ln\frac{0.472\, r_e}{r_w} + s + Dq\right]} . \quad (12)$$

In the preceding equations, s is the familiar skin effect, and (Dq) is an approximate non-Darcy flow term.[4] The non-Darcy flow coefficient D is inversely proportional to gas viscosity, and slightly dependent on time for *very* short transient flows. However, D can be considered constant as an adequate engineering approximation.

APPLICATION OF REAL GAS FLOW EQUATIONS

The following describes application of the real gas flow equations to common cases of gas well flowing testing (drawdown, build-up and back-pressure testing).

DRAWDOWN TESTING

Eqs. 10 or 11 provide the basis for drawdown test analysis for real gas flow. From Eq. 10, it is apparent that a plot of $m(p_{wf})$ vs the logarithm of the producing time for constant rate production should produce a straight line of slope:

$$-b = 5.792 \times 10^4 \frac{qp_{sc}T}{khT_{sc}} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (13)$$

Or for standard conditions of 14.7 psia and 60F:

$$-b = 1{,}637\frac{qT}{kh} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (14)$$

Eq. 14 is analogous to the equation for analyzing build-up and drawdowns for gas wells presented by Tracy,[5] except neither viscosity nor z appear. The striking feature of all of the real gas flow equations presented in the previous section is that pressure-dependent viscosity and gas law deviation factor are absent.

Procedure is as follows. Drawdown data are taken as in the past: measure sandface producing pressures as a function of producing time for constant rate of production. Next, produce a working plot of $m(p)$, psi^2/cps vs pressure in psi for the reservoir. If specific viscosity and density data are available, this can be done by graphical integration or the Trapezoidal rule using Eq. 2. If viscosity and z-factor correlations are normally used, $m(p)$ can be found without integration by reading either Table 1 or Fig. 2, Ref. 1. Drawdown pressures are converted to appropriate $m(p)$ values and plotted vs the logarithm of drawdown time, and a straight line is passed through the data. Either Eqs. 13 or 14 can then be used to find the flow capacity. An example calulation is given in the Appendix.

In general, production of an $m(p)$ vs p working plot for a given reservoir requires less effort than squaring pressures as in current methods. The plot, of course, may be used for all wells producing from the same reservoir if formation temperature and gas composition does not vary widely.

Eqs. 10 or 11 can also be solved for the skin effect and non-Darcy flow coefficient. This result is:

$$s' = s + Dq = 1.151\left[\frac{m(p_i) - m(p_{1\,hr})}{-b} - \log_{10}\frac{k}{\phi(\mu c_t)_i r_w^2} + 3.23\right] \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (15)$$

It is necessary to have drawdown tests at two flow rates to determine both the skin effects and the non-Darcy flow coefficient D separately. As described by Ramey,[6] wellbore storage effects or any variable-rate test can often be used to separate the skin effect and non-Darcy flow. The real gas potential drop across the skin can be found from:

$$\triangle m(p)_{skin} = 0.87\,(-b)s \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (16)$$

Flow efficiency can be found by means of Eqs. 12 or 13:

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$$FE = J_{\text{act.}}/J_{\text{theo.}} = \frac{m(\overline{p}) - m(p_{wf}) - \triangle m(p)_{\text{skin}}}{m(\overline{p}) - m(p_{wf})}. \quad (17)$$

in analogy with the expression presented by Matthews.[7]

The preceding for drawdown analysis for real gas flow should be very accurate. In many cases, conventional plotting of pressure or pressure squared leads to plots which are not linear, although the $m(p)$ plot is linear ~ as it should be. One example given in the Appendix yields reasonable straight lines for both $m(p)$ and pressure-squared plotting (Fig. 2), although a plot of drawdown pressure vs log time has considerable curvature. In other cases, it has been observed that the pressure-log time plot and $m(p)$-log time plots are excellent straight lines, although a pressure-squared plot shows considerable curvature. In all cases tried to date, the $m(p)$-log time plot has produced reasonable straight lines.

Often, it is necessary to analyze a variable-rate drawdown—such as Russell's[8] two-rate test, or to account for wellbore storage effects.[6] This requires superposition of constant rates to generate variable-rate cases. Ref. 1 indicates that superposition can be used as a satisfactory approximation in many cases.

An important question at this point is whether the real gas flow solutions offer a worthwhile improvement over existing gas flow solutions currently used in well test analysis?

In addition to the problem of gas physical property variation, use of the real gas flow solutions offers an important improvement over existing gas flow theory that is particularly important for low permeability formations. That is, the solution in terms of real gas pseudo-pressure includes the second-degree pressure gradient term neglected by methods based on either liquid flow or ideal gas flow solutions. In the case of low-permeability gas well testing, current well test interpretation methods can lead to formation capacity or deliverability estimates that can be grossly in error. This can be illustrated by comparison of flow capacity determinations by various methods for a known capacity system. Excellent data for testing various methods are available from Carter's[9] finite difference solutions for real gas flow.

Carter proposed a method to find flow capacity which was based on a correlation with a dimensionless time wherein viscosity and compressibility were point time functions. Thus, it is possible to compare flow capacities developed from his method, the liquid case analogy and from the real gas flow analysis of this paper. Table 1 presents the comparison. As can be seen, Carter's analysis leads to flow capacities which are high, the liquid flow analogy leads to flow capacities which are generally low, while the real gas flow analysis based on $m(p)$ values leads to capacity estimates in very close agreement with the actual values. Note particularly the good agreement for the Solution 3 case for 8 md-ft capacity. The difference between Carter's analysis and the liquid case analogy is a general result. As flow approaches that of an ideal gas, the two methods and the real gas flow method yields the same answer.

TABLE 1—COMPARISON OF FLOW CAPACITIES ESTIMATED FROM DRAWDOWN DATA BY VARIOUS METHODS FOR CARTER'S GAS FLOW SOLUTIONS (REF. 9)

Solution No.	Flow Rate (Mcf/D)	Initial Pressure (psia)	Capacity (md-ft) True	Carter's	Liquid Case	$m(p)$ Analyses*
1	2,800	3,300	50	53	45.9	49.4
2	4,000	3,300	50	55.3	44.4	50.3
3	1,200	4,700	8	10.92	5.64	8.2

*The differences between the true and real gas $m(p)$ results are not believed significant. Differences are less than normal error involved in taking slopes of drawdown curves.

BUILD-UP TESTING

The pertinent equations for analysis of pressure build-up in terms of the real gas pseudo-pressure are:

$$\text{Flow capacity} = m(p_{w\delta}) = m(p_i) - 5.792 \times 10^4 \frac{qp_{sc}T}{khT_{sc}} \left[\log_{10}\frac{t_p+t_\delta}{t_\delta}\right] \quad (18)$$

Or using standard conditions of 14.7 psia and 60F:

$$m(p_{w\delta}) = m(p_i) - 1{,}637\frac{qT}{kh}\left[\log_{10}\frac{t_p+t_\delta}{t_\delta}\right] \quad (19)$$

Thus, a Horner-type[7] plot of $m(p_{w\delta})$ should yield a straight line of slope $(-b)$, and formation flow capacity can be determined from Eqs. 13 or 14. The skin effect and non-Darcy flow coefficient can be determined from Eq. 15 if the real gas potential difference is replaced by $m(p_{1\text{ hr}}) = (p_{wp})$. The flow efficiency can be determined from Eq. 17 if $m(p_{wp})$ is substituted for $m(p_{wf})$. The pressure drop across the skin can be computed from Eq. 16 as for drawdown.

In a completely analogous manner to previous build-up theory, Eqs. 18 and 19 indicate that the plot of $m(p_{w\delta})$ can be extrapolated to infinite shut-in time (a time ratio of unity) to yield $m(p^*)$. The Matthews-Brons-Hazebroek[10] pressure correction charts can be used to correct the extrapolated $m(p^*)$ value to $m(\overline{p})$, the average real gas potential, if the ordinates of their pressure correction charts are changed to:

$$2.303\left[\frac{m(p^*) - m(\overline{p})}{-b}\right], \quad (20)$$

and the dimensionless producing time is defined as:

$$(t_p)_D = \frac{0.000264\, kt_p}{\phi(\mu c_t)_i A}, \quad (21)$$

where A is the drainage area in square feet.

If producing time t is long enough that the drainage radius is stabilized before the well is shut in, the best approximation for pressure build-up is:

$$m(p_{w\delta}) = m(\overline{p}) - 5.792 \times 10^4 \frac{qp_{sc}T}{khT_{sc}}\left[\log_{10}\left(\frac{\phi(\mu c_t)_i r_e^2}{0.00266\, kt_\delta}\right)\right] \quad (22)$$

The slope of a plot of $m(p_{w\delta})$ vs $\log_{10} t_\delta$ yields the flow capacity, while the $m(p)$ can be read on the straight line at a build-up time t_δ of:

$$t_\delta \text{ hours} = \frac{\phi(\mu c_t)_i r_e^2}{0.00266k}. \quad (23)$$

This is analogous to the Dietz[11] method for liquid flow.

BACK-PRESSURE TEST ANALYSIS

Application of the preceding to back-pressure test analysis is straightforward, but several features deserve mention. First, Eq. 4 provides an excellent match to both short-time isochronal flow tests and stabilized flow tests. The skin effect and the non-Darcy flow constant can be obtained from short-time isochronal flow testing if a single drawdown or build-up is available to obtain the

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flow capacity kh. Once this is done, r_d can be placed equal to 0.472 r_e, and either Eqs. 4 or 12 used to generate a plot of $\log_{10} [m(\bar{p}) - m(p_{wf})]$ vs $\log_{10} q$.

This is analogous to the familiar back-pressure curve. The $m(p)$ curve has many of the characteristics of a normal back-pressure curve, but there are some important differences, also. At low rates, the non-Darcy term Dq will be negligible and the slope of the curve will be unity. At high values of flow rate, non-Darcy flow may become important and the slope of the curve will approach 2 (n value of 0.5) in a fashion similar to that described by Carter *et al.*[12] However, the $m(p)$ back-pressure curve is only slightly sensitive to static pressure level, because only the constant D depends upon pressure through dependence upon gas viscosity (Swift and Kiel[1]). Thus, a single, stabilized back-pressure curve in terms of the real gas pseudo-pressure difference can be used to generate an entire family of back-pressure curves in terms of pressure squared—or used in terms of $m(p)$ with the gas materials balance to provide a simple means of forecasting gas well performance.

AVERAGE GAS PROPERTIES

Current engineering practices in the analysis of gas well tests involves evaluation of an equivalent liquid flow system (Matthews[7]), or modification of expressions for flow of an ideal gas (Tracy[5] or Carter[9]). Matthews recommended that gas properties be evaluated at an arithmetic average pressure, while Carter found empirically that evaluation of physical properties at an average pressure equal to the square root of the average of squared pressures gave reasonable agreement with real gas flow solutions. One important result of the present study is that use of the real gas $m(p)$ leads to a solution for the proper average value for gas physical properties. For example, it is not uncommon to assume that gas vsicosity and gas law deviation factor can be regarded as constant at some average value for the flow region. For a bounded system, this would lead to the flow equation:

$$q = 1.987 \times 10^{-5} \frac{khT_{sc}(\bar{p}^2 - p_{wf}^2)}{p_{sc}T(\mu z)_{avg} \ln \dfrac{0.472 r_e}{r_w}}, \quad . \quad (24)$$

where we have neglected skin effect and non-Darcy flow. If Eq. 24 is compared with Eq. 12, the result is:

$$\frac{\bar{p}^2 - p_{wf}^2}{(\mu z)_{avg}} = m(\bar{p}) - m(p_{wf}) \quad . \quad . \quad . \quad . \quad (25)$$

or

$$(\mu z)_{avg} = \frac{\bar{p}^2 - p_{wf}^2}{m(\bar{p}) - m(p_{wf})} \quad . \quad . \quad . \quad . \quad (26)$$

A similar result will hold if pressures are expressed in terms of p_e rather than $\bar{p}$. Carter[9] suggested the use of an equation similar to Eq. 26 for determination of average properties. Since many gas reservoir flow tests can be analyzed by current theory, Eq. 26 may be of much use. Note however that the $(\mu z)_{avg}$ would be used only in determination of flow capacity from the usual equation—*not* in determination of dimensionless times. Dimensionless times should still be evaluated with physical properties at the initial formation pressure before production.

If Matthew's liquid case analogy is used, an average value of (μB_g) is required for flow capacity determination. A liquid flow analogy to Eq. 12 is:

$$q = 1.987 \times 10^{-5} \frac{kh(2)(\bar{p} - p_{wf})}{\mu B_g [\ln 0.472 r_e/r_w]} \quad . \quad . \quad (27)$$

Comparison of Eqs. 12 and 27 leads to:

$$\frac{2(\bar{p} - p_{wf})}{(\mu B_g)_{avg}} = \frac{T_{sc}[m(\bar{p}) - m(p_{wf})]}{p_{sc}T}, \quad . \quad . \quad . \quad . \quad . \quad (28)$$

or

$$(\mu B_g)_{avg} = \frac{2p_{sc}T(\bar{p} - p_{wf})}{T_{sc}[m(\bar{p}) - m(p_{wf})]} \quad . \quad . \quad . \quad . \quad . \quad . \quad (29)$$

Again, the average value of viscosity-formation volume factor product would be used in determination of flow capacity, not in determination of dimensionless times. Extension of the averages to the transient period is obvious in view of Eq. 4.

It should be emphasized that the liquid case approximation recommended by Matthews, and Tracy's method of gas well build-up analysis, will give excellent results in many cases where pressure drawdown is not large and permeability is high enough that second-degree pressure gradients are not important. Nevertheless, it is recommended that a real gas flow analysis is useful, even in this case, to determine that the current approximate methods are applicable. In any event, it is likely that the real gas flow methods outlined in this paper will be found simpler in application than older methods once a master plot of $m(p)$ vs p is prepared for any given reservoir. This results mainly because it is no longer necessary to square pressures for plotting, and gas physical properties appear only in dimensionless times. An additional benefit is that gas properties are always evaluated at known pressures in the real gas flow method.

CONCLUSIONS

1. As a result of a change of variable called the real gas pseudo-pressure, it is possible to write approximate solutions for the production of real gases from ideal radial systems, which are analogous to the Aronofsky-Jenkins ideal gas flow solutions.

2. The approximate solutions for transient flow of a real gas using the real gas pseudo-pressure leads to methods for interpreting pressure drawdown and build-up tests which are similar to current methods, except that gas physical properties either do not appear in the engineering equations or appear at known pressures. Use of the real gas flow analysis indicates there are many well tests which can be analyzed with current procedures, but there are likely to be many others where current procedures are in error.

3. The real gas flow approximation can be used to establish proper average values of physical properties for current procedures.

Finally, it is emphasized that the real gas pseudo-pressure concept is principally a computing device. Results of back-pressure, build-up and drawdown tests can be converted from $m(p)$ solutions to solutions in terms of pressure or pressure squared to provide familiar information and displays.

NOMENCLATURE

A = drainage area, sq ft

B = formation volume factor, res vol/std vol

b = slope of build-up or drawdown plot, psi²/cp-cycle

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c = compressibility psi^{-1}
D = non-Darcy flow constant, $(Mscf/D)^{-1}$
G_p = cumulative gas produced, scf
h = net formation thickness
J = productivity index, (Mscf/D)/unit potential difference
k = effective permeability
$m(p)$ = real gas pseudo-pressure (Eq. 2) psi^2/cp
p = pressure, psi
$p_D(t_D)$ = van Everdingen-Hurst dimensionless pressure drop
q = gas rate, Mscf/D
r = radial location
s = van Everdingen-Hurst skin effect, dimensionless
s' = apparent skin effect, dimensionless
S_g = fractional gas saturation
T = temperature, °R
t = time, hours
z = real gas law deviation factor, dimensionless ($pv=nzRT$)
ϕ = total porosity, fraction pore volume
μ = viscosity of gas, cp

SUBSCRIPTS AND SUPERSCRIPTS

D = dimensionless
d = drainage
e = exterior boundary
g = gas
i = initial
1 hr = one hour on straight line
m = base
$-$ = superbar, average
sc = standard conditions of pressure and temperature
sf = sand face
t = total
w = well, inner boundary
wf = well flowing
wp = producing before shut-in
$w\delta$ = shut-in well
$*$ = extrapolated
avg = average

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APPENDIX

DRAWDOWN ANALYSIS EXAMPLE

An isochronal flow test is performed on a gas well at two different rates. Given the reservoir data and fluid properties below, determine the flow capacity, skin effect and non-Darcy flow coefficient for this well. The well is completed with tubing-annulus packer.

Reservoir and Gas Data

p_i = 2,300 psia
h = 10 ft
r_w = 0.5 ft
r_e = 2,980 ft (640-acre spacing)
T = 130F
ϕ = 0.1 *fr BV*
S_g = 77 per cent PV

Gas Properties

p(*psia*)	z	Viscosity (cp)
400	0.95	0.0117
800	0.90	0.0125
1,200	0.86	0.0132
1,600	0.81	0.0146
2,000	0.80	0.0163
2,400	0.81	0.0180

Drawdown Data

Flowing Time (hours)	Flow No. 1 (q=1,600 Mscf/D) p_{wf} (psia)	Flow No. 2 (q=3,200 Mscf/D) p_{wf} (psia)
0.232	1,855	1,105
0.4	1,836	1,020
0.6	1,814	954
0.8	1,806	906
1.0	1,797	860
2.0	1,758	700
4.0	1,723	539
6.0	1,703	387

SOLUTION

It can be assumed that wellbore storage effects are negligible, since the well is completed with a down-hole packer. The first step is to find $m(p)$ vs p for this gas.

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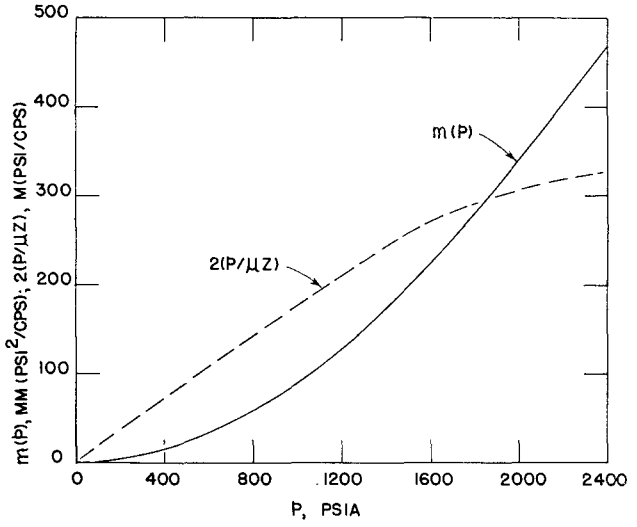

FIG. 1—$m(p)$ AND $2(p/\mu z)$ VS p FOR SAMPLE DRAWDOWN PROBLEM.

This can be done with the gas properties tabulated above and Eq. 2 in the main text. The quantity $2(p/\mu z)$ can be calculated and plotted vs pressure, as shown below and on Fig. 1. Integration can be performed in a tabular calculation by reading mid-point values of $2(p/\mu z)$ from the graph and multiplying by $\triangle p$. The computed $m(p)$, psi²/cp, is also shown on Fig. 1. This curve can be used for future tests with this well or other wells producing the same gas at the same formation temperature. Often, only gas gravity is available. In this case $m(p)$ can be found without integration from Ref. 1.

p (psia)	z	μ (cp)	2(p/μz) (psi/cp)	Mean 2(p/μz)	△p (psi)	2(p/μz) (x△p)	m(p) (psi²/cp)
400	0.95	0.0117	71,975	35,988	400	14.4×10^6	14.4×10^6
800	0.90	0.0125	142,222	107,099	400	42.9×10^6	57.3×10^6
1,200	0.86	0.0132	211,416	176,819	400	70.7×10^6	128.0×10^6
1,600	0.81	0.0146	270,590	241,003	400	96.5×10^6	224.5×10^6
2,000	0.80	0.0163	306,748	288,669	400	115.5×10^6	340.0×10^6
2,400	0.81	0.0180	329,218	319,000	400	127.6×10^6	467.6×10^6

In the following, Flow 1 will be analyzed in detail. Results for Flow 2 will also be given to illustrate the importance of a second flow test at a different rate. Fig. 2 presents the drawdown data plotted in the conventional manner, and in terms of the $m(p_{wf})$.

The flow capacity for Flow 1 can be estimated from Eq. 14 in the main text:

$$kh = 1{,}637\frac{qT}{-b} = 1{,}637\frac{(1{,}600)\,(590)}{(32\times10^6)} = 48.4 \text{ md-ft.}$$

The total of skin effect and non-Darcy flow resistance can be estimated from Eq. 15.

$$s' = s + Dq = 1.151\left[\frac{m(p_i) - m(p_{1\,\mathrm{hr}})}{-b} - \log_{10}\left(\frac{k}{\phi(\mu c_t)_i r_w^2}\right) + 3.23\right]$$

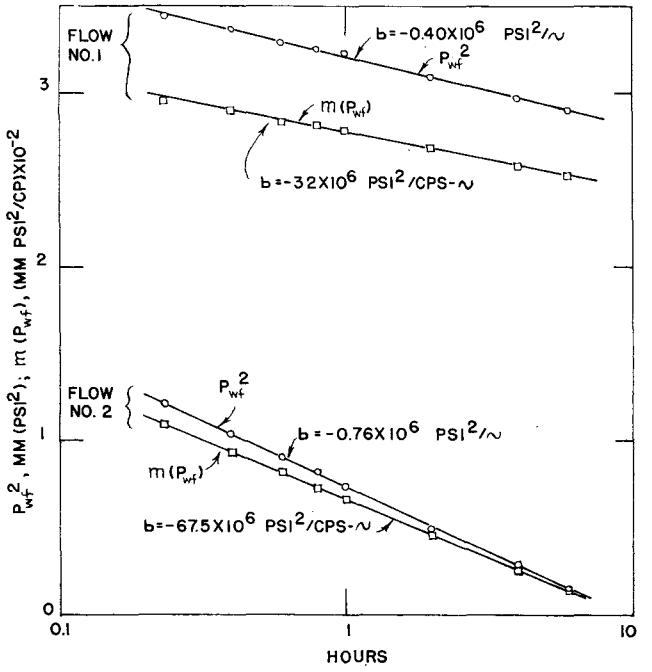

FIG. 2—$(p_{wf})^2$ AND $m(p_{wf})$ VS LOG TIME FOR SAMPLE PROBLEM.

$$s + Dq = 1.151\left[\frac{(435-279)\,10^6}{32\times10^6} - \log_{10}\left(\frac{4.84}{(0.1)\,(0.0176)\,(0.00041)\,(0.77)\,(0.5)^2}\right) + 3.23\right] = 0.657.$$

The flow capacity determined from Flow 2 is 45.8 md-ft, and the skin effect plus non-Darcy flow component is 1.36. Thus

$$\text{Flow 1: } s + Dq_1 = 0.657$$

$$\text{Flow 2: } s + Dq_2 = 1.36.$$

These two equations can be solved for $q_1 = 1{,}600$ Mscf/D, and $q_2 = 3{,}200$ Mscf/D to yield $s = -0.03$ and $D = 4.39\times10^{-4}$ (Mscf/D)$^{-1}$. Thus, the skin effect is negligible and all of the resistance near the well is caused by non-Darcy flow.

The difference in flow capacities found above for the two rates is not significant. This sample problem was taken from two computer solutions by Carter[13] because it provides a good example of an apparent skin effect caused by non-Darcy flow. The true flow capacity used by Carter in the solution was 50 md-ft. The difference between flow capacities determined above and the true value of 50 md-ft results because Carter approximated the effect of non-Darcy flow as a constant pressure-squared difference for his solutions. In real gas flow, a better approximation would be a constant difference in $m(p)$.

The next step would be to substitute the values of kh, s and D into Eq. 12 in the main text to provide a general equation for generation of stabilized deliverability curves. ★★★

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